

Physical

IBR Inverters
GFL/GFM fleet

Line / Busbar
(fault network)

CT / VT
Measurement

Relay

Physical Relay
(Z1, 87L, 67Q, 87B)

Trip Decision
 ≤ 20 ms

Digital Twin

RLS State Estimator
 k_{ibr}, Z_{line}

Scenario Library
N=1000, pre-computed

Protection Model
Mirror of relay logic

Validation

Decision Comparator
DT vs. Relay

Post-Event
Auditor

Model Update
(RLS adapt)

DT cycle: 1 ms (1 kHz) Relay cycle: asynchronous (event-driven)
DT provides 20 pre-fault estimates before fastest Z1/87L trip (20 ms)